

Row-Count Scaling: Federated Simulation Results

How --max-total-rows drove the largest accuracy gains of the whole project

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Executive Summary

federated_simulation.py's data loader caps how many rows survive per (region_cell, attack_type) bucket, auto-sized from a --max-total-rows budget divided across all regions and classes. The default (2,000,000) produces an aggressively small per-bucket cap given 180 real regions -- meaning most of the loaded data was being thinned down hard before it ever reached training. Raising this budget was tested across all three architectures (GRU, CNN, MLP) and produced the single largest, most consistent accuracy gains of the entire project -- larger than any architecture change tested.

GRU's row-count sweep (2M to 10M) moved global accuracy from 74.4% to 86.2% and replay recall from 45.5% to 69.3%. The SAME effect, tested afterward on CNN and MLP at matched low/high row counts, produced comparably large gains on both -- strong evidence this was a data-pipeline fix that helped every architecture, not something specific to sequence models.

Methodological caveat, stated upfront: best-checkpoint restoration (which corrects for a model overfitting past its true peak) was added PARTWAY through this set of experiments. Some runs below reflect final-round results (no correction); others reflect checkpoint-restored results. This is called out explicitly per result rather than silently mixed together -- see the notes under each table.

1. What --max-total-rows Actually Controls

The loader's per-bucket cap is computed as $\text{max_total_rows} / (\text{n_regions} \times 6 \text{ classes})$. With 180 real regions in this dataset, the default 2,000,000-row budget works out to a cap of roughly 1,851 rows per (region, attack_type) combination -- aggressive thinning given some regions have tens of thousands of real rows available. Raising the budget raises this cap proportionally, letting more of each region's real signal survive into training rather than being randomly discarded before the model ever sees it.

--max-total-rows	Auto-sized per-bucket cap	Rows actually sampled
2,000,000 (default)	1,851	1,098,348
5,000,000	4,629	2,221,580
10,000,000	9,259	3,646,553

2. GRU Row-Count Sweep (2M -> 5M -> 10M)

All three runs use window_size=3 and seq_sort_col=lastposupdate (the sequence-ordering fix from earlier testing). Round count was NOT held constant across this sweep (30/30/50) -- the 10M result benefits from both more data AND more rounds, so this table alone cannot cleanly separate the two effects. Section 3 isolates the row-count effect specifically by comparing matched conditions across architectures instead.

Metric	2M rows, 30 rounds	5M rows, 30 rounds	10M rows, 50 rounds
Aircraft dropped (insufficient history)	79.9%	82.6%	84.7%
Clients built	119	129	129
Global accuracy	74.41%	79.93%	86.20%
benign recall	73.04%	80.97%	89.05%
flooding recall	84.75%	88.78%	88.33%
gps_spoof recall	76.87%	78.06%	83.13%
replay recall	45.47%	52.21%	69.28%

Metric	2M rows, 30 rounds	5M rows, 30 rounds	10M rows, 50 rounds
velocity_spoof recall	86.69%	90.19%	93.16%
Total communication	134.52 MB	145.82 MB	243.04 MB

None of these three runs used best-checkpoint restoration (added afterward) -- these are final-round results.

3. Does This Generalize Beyond GRU? Yes -- Tested on CNN and MLP Too

To isolate the row-count effect from GRU-specific factors (sequence windowing, dropout), the same 2M-vs-10M comparison was run on CNN and MLP -- both row-independent architectures with no windowing or dropout concerns at all.

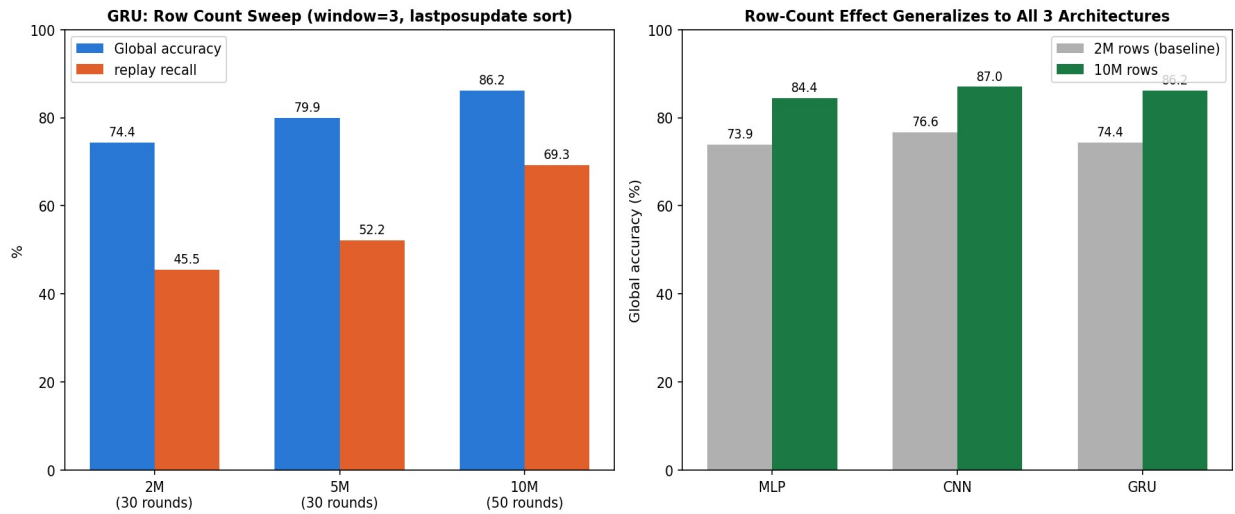


Figure 1. Left: GRU's full row-count sweep. Right: the same low-vs-high row count effect, tested independently on all three architectures.

CNN: 2M vs. 10M rows

Metric	2M rows (15 rounds, no checkpoint)	10M rows (50 rounds, best checkpoint: round 33)
Global accuracy	76.63%	87.01%
benign recall	64.46%	86.19%
replay recall	41.25%	77.48%
gps_spoof recall	80.91%	83.72%
ghost_aircraft recall	99.78%	99.96%
Communication (total / to best checkpoint)	60.03 MB	214.97 MB / 141.88 MB

MLP: 2M vs. 10M rows

Metric	2M rows (15 rounds, no checkpoint)	10M rows (50 rounds, best checkpoint: round 50)
Global accuracy	73.88%	84.44%
benign recall	59.59%	82.33%
replay recall	40.71%	68.61%
gps_spoof recall	78.40%	80.74%

Metric	2M rows (15 rounds, no checkpoint)	10M rows (50 rounds, best checkpoint: round 50)
ghost_aircraft recall	n/a (not recorded in that run)	99.97%
Communication	46.09 MB	165.05 MB

MLP's 10M-row run had NOT yet plateaued at round 50 -- its best checkpoint was the final round itself, train/test gap still small and stable (not overfitting). Unlike CNN, MLP may have further headroom left simply from more rounds at this same row count -- not yet tested.

4. Interpretation

All three architectures gained 8-13 points of global accuracy from the same lever, and replay recall -- the single hardest class all session -- roughly doubled or better in every case (CNN: +36.2 pts, MLP: +27.9 pts, GRU: +23.8 pts from its 2M baseline). This is the strongest, most consistent single finding of the entire project: the data pipeline's row-budget sizing was a bigger bottleneck than any architecture choice tested.

A secondary implication worth stating directly: because this fix is architecture-agnostic, it should be treated as the DEFAULT going forward, not an optional flag to remember -- any future architecture comparison run at the old 2,000,000-row default is not a fair comparison and should be re-run before its numbers are trusted.

5. Limitations and Open Items

- Round counts were not held constant within the GRU sweep (30/30/50) -- the 5M-to-10M jump conflates more data with more training. A clean re-run at matched round counts would isolate the row-count effect precisely.
- Best-checkpoint restoration was added mid-sequence -- the GRU sweep and both 2M baselines predate it and reflect final-round (potentially under- or over-stated) results, while the two 10M CNN/MLP runs benefit from it. A full re-run of the sweep with checkpointing enabled throughout would give a fully consistent set of numbers.
- Not yet tested: whether raising --max-total-rows further (e.g. 20,000,000) continues improving results or has begun to plateau -- per-class row availability suggests diminishing returns may be close, since the rarest attack classes (807,515 rows each across the whole day) are likely already close to fully represented at the 10M cap.
- A wider MLP (--mlp-hidden-sizes) at 10M rows was proposed as a follow-up to test whether MLP's gap to CNN is a capacity limit or an architectural one -- not yet run at the time of this report.